

Local Differentiated-Role Occupancy in the Hilbert Substrate Framework

March 21, 2026

Abstract

This note defines a local witness for *differentiated expression* in the Hilbert Substrate Framework (HSF). The purpose of the witness is to distinguish raw growth in active support count from genuine growth in faithfully differentiated local function. The central object introduced here is *local differentiated-role occupancy*, denoted $\mathcal{O}_{\text{diff}}(R)$, which measures the effective number of functionally distinct local roles carried in a region R . This witness is intended to support lawful move scoring in the mesoscape program, especially the question of when differentiated expression saturates before occupancy saturates. The note gives a formal definition, an interpretable simplified proxy, and the move-level quantities needed for implementation in bookkeeping and scoring code.

1 Motivation

The current mesoscape program distinguishes between:

1. **occupancy**, meaning how many explicit support patches are currently active, and
2. **expression**, meaning how much faithful differentiated local function is being realized.

These are not the same thing. A region may continue to accumulate active support while adding little or no genuinely new local function. In that case, occupancy continues to rise while differentiated expression saturates. If the lawful move law is to support pre-saturation breathing, compression before full box-filling, or weakening/retirement as genuinely lawful channels, then the expression term must respond to *role differentiation* rather than to raw support count.

The desired witness must therefore answer the following question:

How many *functionally distinct* local roles are actually being carried in a region, as opposed to how many support patches are merely present?

This note defines a witness intended to answer that question.

2 Region of evaluation

Let R denote a local comparison region. This region should be chosen in the same spirit as other lawful move comparisons: it may include

- a candidate support patch or interface,

- its immediate neighborhood,
- any overlapping organizer core or shell region,
- any local support whose activity or redundancy is directly affected by the candidate move.

The witness defined below is local by construction. It is not intended to characterize the entire global state in one step. It is intended to say whether the *local active role pattern* in R is still differentiating, or has already become mostly redundant.

3 Role candidates

Let

$$\mathcal{U}_R = \{u_1, \dots, u_m\}$$

be the set of currently active role candidates in the region R .

A role candidate is *not* merely a node or active support patch. Rather, it is a locally active support-bearing contribution to differentiated expression. In practice, a role candidate may correspond to an active patch, shell contributor, interface-adjacent site, or other bookkeeping unit chosen by the implementation, provided that it is intended to represent an actual local functional contribution.

4 Role fingerprints

Each active role candidate $u_a \in \mathcal{U}_R$ is assigned a local fingerprint vector

$$\phi(u_a) = (N_a, R_a, D_a, K_a, B_a).$$

The components are interpreted as follows.

N_a	Novelty contribution. A score measuring whether u_a carries channel-role content that is locally underrepresented.
R_a	Relief contribution. A score measuring whether u_a relieves stress, overload, concentration, or burden on parent/core/shell structure.
D_a	Distinctness contribution. A score measuring whether u_a is functionally distinguishable from nearby siblings or neighboring supports.
K_a	Committed-interface participation. A bookkeeping score indicating whether u_a participates in interfaces that are actually carrying maintained commitments.
B_a	Bandwidth or influence participation. A score or descriptor indicating whether u_a is part of an active influence distribution rather than inert support residue.

These quantities are intentionally local and proxy-friendly. The witness does not require dense reduced-state fidelity, dense mutual information, or global exact-state comparisons in its defining form.

5 Effective activity weights

Not every active role candidate should count equally. A weakly active or barely function-bearing candidate should contribute less to local differentiated-role occupancy than one that is strongly active and clearly participating in the organizer’s functional expression.

We therefore assign each role candidate a nonnegative effective weight

$$w_a \geq 0.$$

A general form is

$$w_a = \alpha_N N_a + \alpha_R R_a + \alpha_D D_a + \alpha_K K_a + \alpha_B B_a,$$

with nonnegative coefficients

$$\alpha_N, \alpha_R, \alpha_D, \alpha_K, \alpha_B \geq 0.$$

If desired, one may enforce nonnegativity explicitly by replacing the right-hand side with $\max(0, \cdot)$.

In the intended use, novelty should not be underweighted. A reasonable policy is that the coefficients satisfy

$$\alpha_N \geq \alpha_R, \alpha_D, \alpha_K, \alpha_B,$$

since channel novelty is the primary marker of genuinely differentiated expression.

6 Role overlap kernel

To distinguish genuinely different roles from repeated copies of the same role, we introduce an overlap kernel

$$K_{ab} = \text{sim}(\phi(u_a), \phi(u_b)), \quad K_{ab} \in [0, 1].$$

This kernel measures the degree to which role candidates u_a and u_b are functionally overlapping:

- $K_{ab} \approx 1$ means they are effectively the same role,
- $K_{ab} \approx 0$ means they are clearly distinct roles.

A concrete weighted form is

$$K_{ab} = \beta_N \text{sim}_N(N_a, N_b) + \beta_R \text{sim}_R(R_a, R_b) + \beta_D \text{sim}_D(D_a, D_b) + \beta_K \text{sim}_K(K_a, K_b) + \beta_B \text{sim}_B(B_a, B_b),$$

where

$$\beta_N, \beta_R, \beta_D, \beta_K, \beta_B \geq 0, \quad \beta_N + \beta_R + \beta_D + \beta_K + \beta_B = 1.$$

As with the activity weights, novelty and distinctness should not be underweighted. The kernel is the anti-cloning mechanism in the definition: if many active supports all carry the same role fingerprint, then they should collapse toward a single effective role rather than be counted independently.

7 Definition of local differentiated-role occupancy

We now define the main witness.

[Local differentiated-role occupancy] Let R be a local comparison region, let $\mathcal{U}_R = \{u_1, \dots, u_m\}$ be its active role candidates, let $w_a \geq 0$ be their effective activity weights, and let $K_{ab} \in [0, 1]$ be their overlap kernel. The *local differentiated-role occupancy* of R is

$$\mathcal{O}_{\text{diff}}(R) = \frac{(\sum_{a=1}^m w_a)^2}{\sum_{a=1}^m \sum_{b=1}^m w_a w_b K_{ab}}.$$

This quantity is an effective-rank or participation-rank measure. It behaves as follows:

- if all active role candidates in R are redundant copies of the same role, then $\mathcal{O}_{\text{diff}}(R) \approx 1$;
- if there are k comparably active, genuinely distinct roles in R , then $\mathcal{O}_{\text{diff}}(R) \approx k$.

Thus $\mathcal{O}_{\text{diff}}(R)$ measures not how many supports are present, but how many *distinct local roles* are effectively being carried.

8 Interpretation

The intended interpretation is straightforward:

Local differentiated-role occupancy is the effective rank of actively carried local support roles in a region, counted by faithful novelty, relief contribution, and functional distinctness rather than by raw support multiplicity.

This makes it possible to distinguish two physically different situations:

1. **Occupancy growth with real differentiation:** new support creates genuinely new local roles, so both active support count and $\mathcal{O}_{\text{diff}}$ rise.
2. **Occupancy growth with redundancy:** new support mostly duplicates roles already present, so active support count rises while $\mathcal{O}_{\text{diff}}$ stays flat or may even decrease.

The second regime is precisely the one in which differentiated expression saturates before occupancy saturates.

9 Saturation criterion

Let $A(R_t)$ denote the raw active-support count in the same local region at time t . Then *pre-saturation differentiated-expression saturation* occurs when

$$\frac{d}{dt}\mathcal{O}_{\text{diff}}(R_t) \approx 0 \quad \text{while} \quad \frac{d}{dt}A(R_t) > 0.$$

That is, local active support continues to grow, but the effective rank of distinct roles does not. This is the central operational criterion:

Differentiated expression saturates before occupancy saturates when local role rank plateaus while raw support count is still increasing.

10 Move-level form

For a candidate lawful move M evaluated in region R , define

$$\Delta\mathcal{O}_{\text{diff}}(M; R) = \mathcal{O}_{\text{diff}}^{\text{after}}(R) - \mathcal{O}_{\text{diff}}^{\text{before}}(R).$$

This quantity provides a natural local expression witness for move scoring.
The intended sign policy is:

- **Birth** is supported when $\Delta\mathcal{O}_{\text{diff}}(M; R) > 0$.
- **Birth** flattens when $\Delta\mathcal{O}_{\text{diff}}(M; R) \approx 0$.
- **Birth** is penalized when $\Delta\mathcal{O}_{\text{diff}}(M; R) < 0$.
- **Weakening** or **retirement** may receive positive expression credit when removing a redundant support increases effective differentiated-role occupancy by cleaning up duplication.

This move-level witness therefore creates a lawful route for:

1. pre-saturation breathing,
2. compression before complete box-filling,
3. weakening as an actual lawful channel rather than a dead option,
4. retirement as removal of locally evacuated redundancy rather than arbitrary shrinkage.

11 Simplified proxy form

For initial implementation, one may use a reduced fingerprint

$$\phi(u_a) = (N_a, R_a, D_a),$$

ignoring K_a and B_a on the first pass.

Then define

$$w_a = \max(0, \gamma_N N_a + \gamma_R R_a + \gamma_D D_a),$$

with

$$\gamma_N, \gamma_R, \gamma_D \geq 0, \quad \gamma_N \geq \gamma_R, \gamma_D.$$

Use a reduced overlap kernel

$$K_{ab} = \gamma_N \text{sim}_N(N_a, N_b) + \gamma_R \text{sim}_R(R_a, R_b) + \gamma_D \text{sim}_D(D_a, D_b).$$

Then compute

$$\mathcal{O}_{\text{diff}}(R) = \frac{(\sum_{a=1}^m w_a)^2}{\sum_{a=1}^m \sum_{b=1}^m w_a w_b K_{ab}}.$$

This simplified form is likely sufficient for the first bookkeeping implementation.

12 Practical interpretation for the mesoscape program

The witness $\mathcal{O}_{\text{diff}}(R)$ is designed to diagnose a specific failure mode:

support count outruns role differentiation.

If the current implementation is still allowing births to gain too much credit merely for adding support, then one expects to see:

- active support count continuing to increase,
- $\mathcal{O}_{\text{diff}}(R)$ flattening earlier,
- weakening or retirement still losing because redundancy is not being penalized strongly enough.

In that case, the issue is not that the substrate “wants” more support in every circumstance. The issue is that the expression bookkeeping is still too permissive toward duplication.

Conversely, if both active support count and $\mathcal{O}_{\text{diff}}(R)$ continue to rise together almost to capacity, then the finite-box exact regime may truly favor broad local support expansion before compression becomes competitive.

Thus the witness is not merely a scoring ingredient. It is also a diagnostic object for determining which scientific interpretation is actually supported by the current engine.

13 Minimal implementation interface

A practical first implementation may expose the following move-local diagnostics:

$$A_{\text{before}}(R), \quad A_{\text{after}}(R), \quad \mathcal{O}_{\text{diff}}^{\text{before}}(R), \quad \mathcal{O}_{\text{diff}}^{\text{after}}(R), \quad \Delta\mathcal{O}_{\text{diff}}(M; R).$$

In addition, each role candidate u_a should expose a compact loggable fingerprint, for example:

$$(N_a, R_a, D_a) \quad \text{or} \quad (N_a, R_a, D_a, K_a, B_a).$$

This would allow post-run analysis scripts to determine:

1. when local role rank plateaus,
2. whether births after that point are mostly redundant,
3. whether retirement and weakening are eliminating truly evacuated roles or merely pruning active structure,
4. whether pre-saturation breathing is present.

14 Conclusion

The central point of this note is simple:

Differentiated expression should be tracked by the effective number of functionally distinct local roles, not by raw support count.

The proposed witness for this is local differentiated-role occupancy,

$$\mathcal{O}_{\text{diff}}(R),$$

defined as an effective-rank measure on local active role candidates weighted by functional activity and collapsed by functional overlap.

This provides:

- a principled local witness for expression saturation,
- a direct move-level quantity $\Delta\mathcal{O}_{\text{diff}}(M; R)$,
- a route to lawful pre-saturation compression,
- a way to distinguish real differentiated growth from redundant support accumulation.

In the mesoscape program, the decisive empirical question is therefore not merely whether occupancy grows, but whether local differentiated-role occupancy grows with it. If occupancy rises while $\mathcal{O}_{\text{diff}}$ plateaus, then differentiated expression has saturated before support occupancy has, and the lawful move system should begin rewarding redundancy removal rather than continued local duplication.